Deep Learning in Smart Applications: Approaches and Challenges

M. Sowmiya, B. Banu Rekha, and R. Kanthavel

1 Introduction

The concept of smart applications is rising rapidly due to an increase in urbanization of developed and developing countries. The main goal of smart applications is to minimize human intervention and maximize the use of automated technologies. In the deployment of sensors, wearable devices, and actuators in various domains, a massive amount of data is generated at high speed. Also, smart cities are deploying intelligent technologies to automate the process. Mainly, in sectors such as transportation, environmental, healthcare, and agriculture, industries are currently presenting artificial intelligence (AI) techniques. Rapid developments in AI provide the platform to process the huge amount of data, analyze the pattern, and forecast the outcome [1]. This paper discusses the progress of deep learning (DL) methods in environmental and healthcare sectors. Specifically, this paper presents a review of the theoretical background, deep learning concepts, methodology involved in waste management, and retinal image analysis of diabetic retinopathy. Figure 1 highlights the era of using deep learning in smart city development.

The main contributions of this work focus on the study of deep learning in two fields which are described as follows:

1. Waste segregation is a challenging part of making the environment smarter. For developing a sustainable economy, it is essential to use intelligent systems that

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R. Maheswar et al. (eds.), *Challenges and Solutions for Sustainable Smart City Development*, EAI/Springer Innovations in Communication and Computing,
https://doi.org/10.1007/978-3-030-70183-3_3

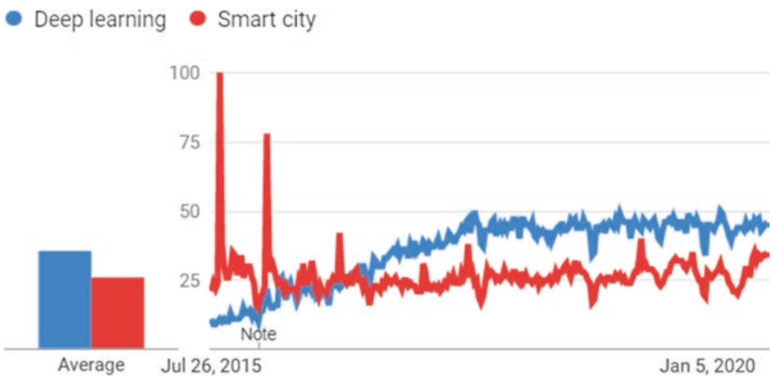


Fig. 1 Popularity of deep learning and smart city over the past 5 years (Source: Google trends)

require less manpower. The smart bins are emerging to segregate the waste and put it into corresponding waste categories. In recent days, deep learning algorithms have been used to automatically segregate the recycling waste to develop the prototype. This paper highlights the use of convolutional neural network (CNN) to classify and identify the properties of waste so that it can be employed in smart bins.

2. Artificial intelligence techniques applied in the healthcare domain allow the professionals to monitor, diagnose, and highlight the province of the problem and propose the possible required solution. Moreover, many smart devices are emerging to reduce the cost of public healthcare systems. Recently, smart contact lenses to diagnose and treat diabetic retinopathy [2] and smartphone-based DR detection systems have been developed [3]. Deep learning algorithms have rapidly turned out to be a methodology for analyzing and diagnosing the various medical pathological conditions. This paper provides the knowledge of abnormalities, lesions identified in retinal image to diagnose diabetic retinopathy (DR), and DL methods used to segment the features and identifies the issues in diagnosis.

We also discuss the challenges in current research and provide the suggestions for future researchers.

2 Waste Segregation and Classification

Waste management is one of the challenging issues for both developed and developing countries due to the rise in population growth, urbanization, and industrialization. According to the global survey given by the World Bank, waste generation is expected to increase from 2.01 billion tons (2016) to 3.40 billion tons (2050). Urban people in India generate 62 million tons of municipal solid waste (MSW) each year, and only 11.9 million tons is treated for disposal and recycling [DownToEarth survey]. Wastes are often dumped in landfill sites or openly burned. These uncollected waste practices create the effect of health issues and environmental consequences like climate change and air and water pollution. On making the sustainable cities, municipalities focus on an integrated approach based on the 3Rs (reduce, reuse, and recycle) [4]. However, few efforts have been made to recycle the amount of waste generation at source due to the limitation of resources.

Waste management is a shared activity of segregation, collection, transportation, recycling, and disposal of waste [5]. Waste segregation and recycling is an essential component for building a smart garbage system. Segregation of solid waste requires manpower, which brings many health issues for the waste sorters. Recently due to the COVID-19 situation, people are disposing of gloves and masks mixed with other wastes and making the waste collectors lives at risk. Additionally, this is a time-consuming and less efficient process due to the massive amount of waste generated each day. Recent advancements in technology have given the automated approach of segregating the waste from households to industries. Researchers have been exploring automated waste segregation techniques to advance the proficiency of the recycling process. This paper provides a comprehensive review of recent research on the segregation of recyclable wastes using artificial intelligence techniques. This paper focuses on DL methods to identify and classify the waste from the image acquisition system, which can be further reused or recycled. Also, we discussed the widely used DL architectures, available datasets, and challenges yet to be explored in the future with fruitful suggestions.

2.1 Waste Materials

Municipal solid waste (MSW) includes mostly recyclable waste (paper, plastic, glass, metals, etc.) and nonrecyclable wastes (food waste, medical waste, sanitation waste) [EPA, US]. From the survey given by the US Environmental Agency, recycling progression of materials like plastics, metals, and glass is limited as shown in Fig. 2. Recyclable waste can be processed to produce useful products, and recycling reduces the amount of waste sent to landfills and incinerators. This review work will primarily focus on the convergence of DL techniques in image-based recyclable waste identification systems, which can be implemented initially in the automated process to segregate the wastes.

2.2 Database Information

The main aim of the study discusses the deep learning models to classify the recyclable wastes like paper, plastic, metal, glass, and others. But there are only a few datasets available for the research openly. Waste image datasets comprise of either collected from Google search, acquisition systems, or existing open image

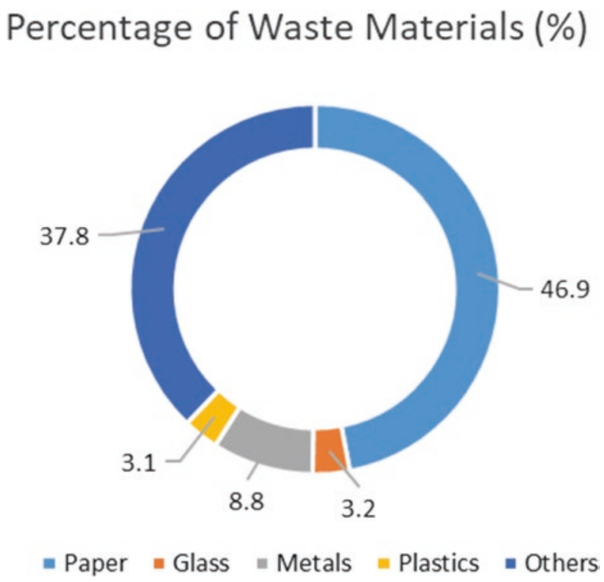


Fig. 2 Recycling and composting proportion of materials (Source: United States Environmental Protection Agency, 2017)

Table 1 Examples of waste image categories

Waste class	Waste items	Recyclable/others
Paper	Cups, books, newspaper, cardboard	Recyclable
Plastic	Shampoo bottles, bags	Recyclable
Metal	Can, pen, cap	Recyclable
Glass	Bottles	Recyclable
Fruit/vegetable	Apple, orange, carrot, onion	Others
Kitchen waste	Lunch box, egg, cooked rice	Others
Trash	Other items	Others

databases. There are few open databases available such as TrashNet [6], Flickr Material Database (FMD) [7], and Materials in Context Database (MINC-2500) [8].

Our focus is on image-based recyclable waste identification, and most of the researchers used the TrashNet dataset for related works. TrashNet contains a collection of 2527 images from 6 categories, paper, plastic, metal, glass, cardboard, and trash. The trash images are taken from the Stanford area with white background and lighting variations using mobile devices. Each picture is resized down to a spatial resolution of 512 × 384 pixels because of the resource limitations. Researchers also created their own dataset with specific categories. Table 1 shows a few examples of waste classes and their subgroup. Waste images can be generated through an image acquisition system or through smartphones. The camera is connected to a controller like raspberry pi 3 which implements a Python script that captures images [9] of waste types, to achieve better knowledge during training.

Fig. 3 Example of waste images

Fig. 4 Example of metal cans with different shapes

Figure 3 shows a sample of recyclable waste categories. The waste image has slight variations than the original image, due to change in shape and properties of waste as a result of being thrown away. As shown in Fig. 4, the same metal can have three different shapes. Therefore, precise classification and identification of the above said cases are challenging ones. Deep learning algorithms are able to extract those features and accurately identify waste types.

2.3 Methodology in Waste Segregation

Waste segregation using computer vision techniques involves obtaining images from the camera, object detection, feature extraction, and classification as recyclable or not. Since the real-time image is noisy and random, pre-processing the data will improve the performance of the model. Various pre-processing steps such as image resize, removal of noise, and enhancing the brightness [6] have been employed. Data normalization methods such as *Z*-score and Min-Max can be used if necessary, to make the model less prone to bias.

2.3.1 Object Detection

Multiple waste items and non-waste items might be present in real-time captured garbage images. During this scenario, each waste item has to be identified separately and classified into a particular class. Few kinds of research have addressed object detection in waste segregation so far. Region-based convolutional neural network (R-CNN) models have been used to detect and identify the size of the waste. It may require the size of all the images in the training set to be the same. Authors have developed a model with Faster R-CNN to classify waste into paper, recycling, and landfill from multiple objects in a single image with precision accuracy of 68.3% [10]. Similarly, E-waste can be detected using Faster R-CNN with an accuracy of 90% [11].

2.3.2 Feature Extraction and Classification

Size, shape, texture, and the color of the waste item are the essential features that can be extracted during the training of deep neural models like CNN. Principal component analysis (PCA) might be used to reduce the dimension of the feature vectors prior to the classification stage. The hybrid system can be developed by using the features extracted from the CNN model and process it to a classifier as shown in Fig. 5. Features are extracted using pre-trained models and a multilayer perceptron is trained to identify the waste calss [12]. From the study, we found that SVM along with CNN can improve prediction accuracy [13]. Moreover, the additional information regarding the waste properties can be given by the sensing unit which enhances the CNN prediction [12].

2.3.3 Supervised Architectures for Waste Image Classification

Convolutional neural network (CNN) is one of the most familiar deep learning algorithms that performs sequential operations of convolution and pooling to extract the features from the data. CNN provides the spatial low-dimensional representation with robust features. Applications like image classification, segmentation, and

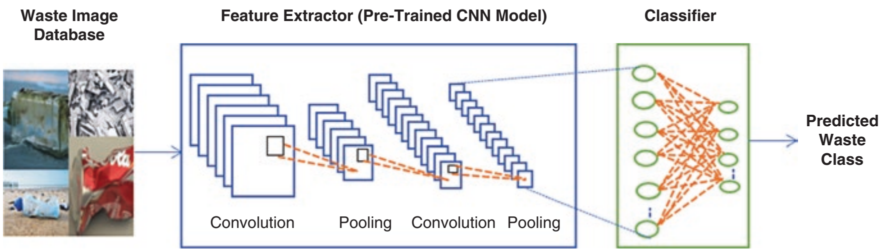


Fig. 5 Predicting the waste class using pre-trained model

detection are extensively using CNN to achieve state-of-the-art results. Many researchers used CNN architectures to perform waste image classification and Faster R-CNN to detect the waste object in the image. Deep architectures, namely, AlexNet, GoogleNet, VGGNet, and ResNet [14], pre-trained on ImageNet are commonly used in the research work. AlexNet is a collection of 8 layers, GoogleNet consists of 22 layers, while VGGNet and ResNet are available in several layer configurations.

2.3.3.1 AlexNet Architecture for Waste Image Classification

AlexNet architecture outperforms the CNN in ImageNet LSVRC-2012 competition with minimal error rate. It consists of eight layers including a convolutional layer and fully connected layers as shown in Fig. 6. To extract the deep features, each layer used kernels such as 11 × 11, 5 × 5, and 3 × 3. After convolution, ReLU improves the convergence followed by the overlapping pooling layer. Finally, 1000 class labels are prophesied by the Softmax layer. To reduce overfitting, data augmentation, and dropout in fully connected layer was implemented [13]. Similarly, other architectures used in waste segregation are detailed in Table 2. Table 3 summarizes the variant of the CNN model instigated for various waste image datasets and highlights the practice used.

AlexNet pre-trained on the ImageNet database can act as a feature extractor. ImageNet database contains 15 million high-resolution images with 22,000

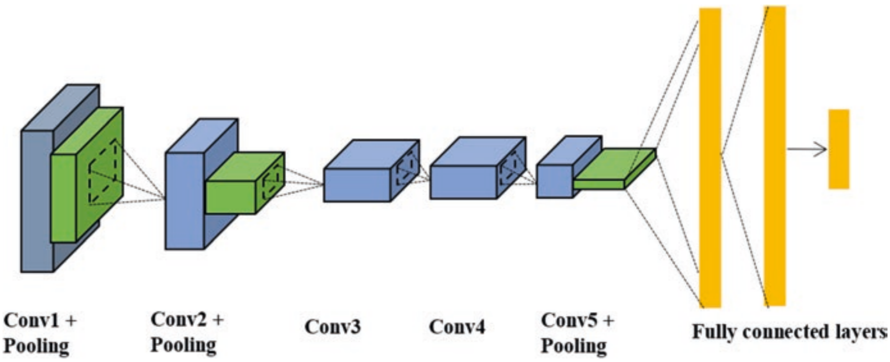


Fig. 6 AlexNet architecture

Table 2 Characteristics of CNN architectures

Architecture	Year	No. of layers	Parameters
AlexNet	2012	8	~60 M
VGG16	2014	16	~138 M
GoogleNet	2014	22	~6 M
ResNet50	2015	50 (residual connections)	~26 M
Inception-v3	2015	48	~23 M

Table 3 Outline of deep learning architectures for waste image classification

References	Architecture	Dataset	Waste item	Observations	Accuracy
[6]	VGG16, ResNet18	TrashNet (1800)+ own (1302)	Paper, plastic, metal, glass	Used data augmentation and transfer learning, overfitting occurred in VGG16, ResNet18 produced the best outcome	87%
[9]	CNN (AlexNet) and SVM	Own	Plastic, paper, and metal	Overfitting is reduced	CNN—83% SVM—94.8%
[12]	CNN (AlexNet) + multilayer perceptron (MLP)	Own	Recyclable (paper, plastic, metal, glass) and nonrecyclable (fruits/ vegetables, kitchen waste)	Data augmentation, identifying multiple wastes in images using Faster R-CNN	90%
[14]	AlexNet, VGG16, GoogleNet, and ResNet	Own (garbage images)	2527 images with 6 different classes	Classifier—Softmax, SVM	GoogleNet + SVM—97.86%
[15]	VGG, Inception, ResNet	TrashNet	Paper, plastic, metal, glass, cardboard, general trash	Inception—ResNet produced the best performance	88.6%
[16]	Inception-ResNetV2, DenseNet121	TrashNet	Paper, plastic, metal, glass, cardboard, trash	Transfer learning, fine-tuning the parameters, optimizers like adam and adadelta are used	InceptionResNetV2—87% DenseNet121—95%
[17]	CNN	Garbage In Images (GINI)		Reduction in prediction time, memory usage	87.69%

categories [13]. Waste datasets are tested with the pre-trained model to classify the waste items.

Comparison of VGG, Inception, and ResNet models has been analyzed for the TrashNet dataset. From the analysis, combining Inception-ResNet models produced more accuracy than the individual model [15]. In addition, it is found that some models are biased toward particular waste item due to imbalance in data categories; low inter-class variations like the plastic bottle are confused with a glass bottle. Development of diverse architectures such as GarbNet [18] or OscarNet [18], which are based on pre-trained CNN architectures such as AlexNet or VGG-19, has been made for waste classification. However, researchers choose to train the model on a massive dataset like ImageNet and then extract the features and used it for image classification.

2.3.3.2 Faster R-CNN

Faster R-CNN is a variant of the R-CNN family developed in 2015. It is widely used in object detection applications, where the bounding box for each object is generated and features of those objects are extracted using backbone CNN. Faster R-CNN is implemented to detect the objects in food tray images along with CNN that produced the mean average precision of 86% [19], as shown in Fig. 7. It can be employed in a multi-waste image to abstract the single waste and fed for further process.

2.4 Network Training Methods

2.4.1 Transfer Learning

In applications, research using transfer learning has gained more attention due to limited computational resources and a lack of large-scale datasets for training. However, the number of images used to train the deep CNN from scratch is very less due to limited availability of open-source waste dataset. Identifying an accurate waste classification model is a critical task because of the varying nature of wastes and their method of disposal. Most of the researchers have used a pre-trained network for identifying wastes. The two approaches used for training and classification are:

1. Using a pre-trained network as a feature extractor and fine-tuning a pre-trained network on waste data
2. Training a CNN model from scratch using the available databases

Pre-training a model using large datasets like ImageNet [13] and features are extracted, fine-tuning the model with the specific application data. Since, pre-training a network makes the model to learn the parameters, so that it can be used for any similar applications.

Fig. 7 Example for bounding box for each object in food tray [19]

2.4.2 Data Augmentation

Overfitting is the challenging issue confronted in deep learning methods for image-based applications. Deep learning models produce accurate results based on the training set. For some applications, due to a smaller number of images available for training, the overfitting phenomena will arise. To overcome the above issue, data augmentation has been developed to deep neural networks on smaller datasets [20]. Image augmentation methods like rotate, crop, horizontal flip, and the vertical flip are commonly used to generate a greater number of data for training. Figure 8 shows the sample augmented images.

During the validation phase, *k*-fold cross-validation method can be employed to evaluate the performance of the model before testing. Cross-validation can effectively improve the generalization ability of the model and overcome the overfitting [21]. For example, using tenfold cross-validation, divide all datasets into ten copies, take one copy each time for testing and take nine copies for training, and finally obtain the average.

2.5 Challenges and Future Research Ideas

In this section, the major research gap identified from the existing work for segregation of waste categories and suggestions the researcher can take up to develop their ideas. From the survey, it is understood that there are still many challenges ahead for this developing area owing to the limited datasets, real-time constraints, and complex nature of deep learning:

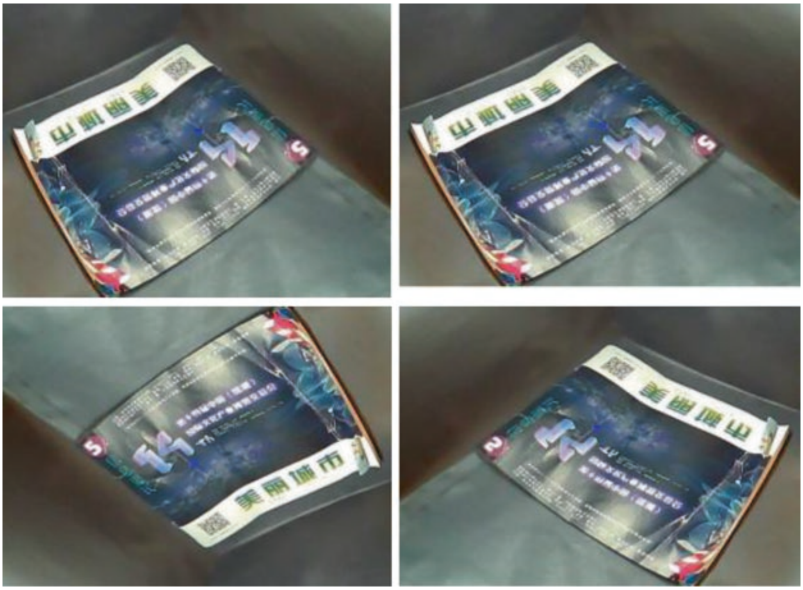


Fig. 8 Example of flipped augmented images

1. A limited number of data reduce the training accuracy and cause overfitting. Techniques like transfer learning, data augmentation, and dropout can improve the performance on smaller datasets.
2. Imbalanced dataset, i.e., some of the classes in the dataset have a larger number of images which make the model to become unfair. The uneven distribution of data will affect the model.
3. Fine-tuning the hyperparameters like learning rate, epochs, and optimizer helps in achieving better testing accuracy.
4. Multiple deep models can achieve an advantage over individual, i.e., individual CNN models can be trained and features are extracted to form a new feature vector. Using fusion techniques [22], the feature vector can be analyzed and classified to improve the overall performance of the waste classification problem. Also, feature vectors can be directly fed to fully connected layers.
5. Ensemble classifiers proved to perform well when compared to single base classifiers [23]. Three types of ensemble methods are bagging, boosting, and stacking. Researchers can implement ensemble methods for waste image identification.

The following are the main constraints in implementing the automatic waste segregation system in real time:

1. High-speed servers and networks are required to manage and process the information from data acquisition unit and sensors.

2. Deployment of the designed model requires high computational resources like memory, power, and cost.
3. Response time of segregating the waste must be minimum, to make use of the waste at an early stage.

3 Diabetic Retinopathy

Diabetic retinopathy (DR) is a threatening complication of diabetics, in which the retina is affected due to an increase in blood sugar level. DR leads to vision loss if not detected and treated early. The primary challenge lies in the early detection of DR because there may be no symptoms that often occur in early-stage and untreatable at advanced stages. Early screening of diabetic patients requires well-trained clinical experts, patient attention, and laborious process. Recent researches are highlighting the need for an automated diagnosis system in healthcare. Retinal image analysis is evolving for the early detection of systemic diseases because of the noninvasive visualization of the retinal vessel structure. Ophthalmologists can detect the pathological and microvascular abnormalities by examining the retinal image of the patients. An automated diagnosis system is emerging to assist experts in early DR diagnosis and ocular diseases.

The objectives of this review for DR screening:

1. To summarize the theoretical background in DR
2. To review the existing works for the detection of retinal lesions using deep learning methods for early diagnosis

From the study, it is found that deep learning is emerging for the detection of retinal lesions and quite a number of researches started using deep learning vigorously.

3.1 Retinal Imaging Modalities

Imaging modalities like Fundus photography and optical coherence tomography (OCT) have been developed for visualizing retinal structures given in Fig. 9. Fundus photography provides a 2D representation, which uses a low-power microscope with an attached camera for imaging the interior surface of the eye which includes the retina, optic disk (OD), retinal vasculature, and macula. Fundus photography is widely used for early screening and diagnosis of diseases such as AMD, glaucoma, and DR [24]. OCT is an advanced imaging modality that uses low coherence interferometry to produce cross-sectional images of the retina. Recently, OCT has been used for the detection of diabetic macular edema [24].

Fundus photography is primarily useful for monitoring the progression of diabetic retinopathy over time. DR is diagnosed by the abnormalities and lesions developed over the retina. This work discusses the deep learning approaches developed for early diagnosis system using the retinal image and reviews the retinal

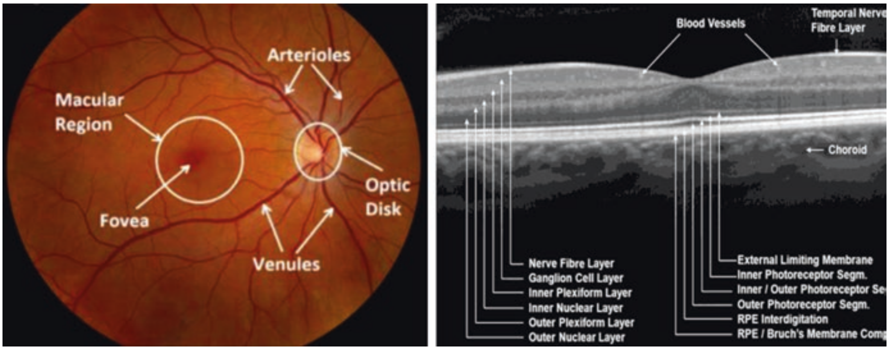


Fig. 9 (**a**) Normal Fundus image. (**b**) Cross-sectional view of retinal layers [Source: Google]

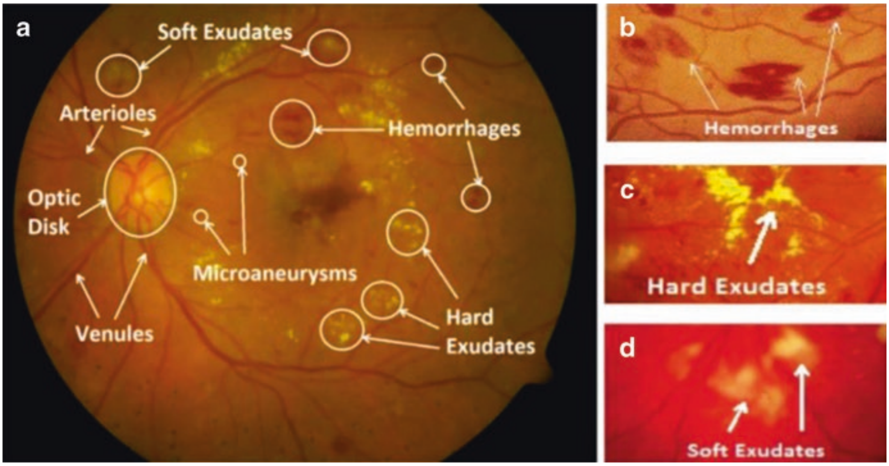


Fig. 10 Fundus photograph showing abnormal lesions and morphologies [24]

abnormalities, various stages of DR, methodology established by the current researches, and challenges faced in DR detection.

3.2 *Retinal Lesions*

DR may cause several pathological lesions in the retina, which are shown in Fig. 10:

Microaneurysms (MAs)—Early symptom of DR, due to small bulges that cause leakage of the retinal vessel. MAs are small circular red spots which are usually less than 125 microns in diameter with sharp margins [24]. DR severity depends on the number of MAs.

Hemorrhages (HM)—Bleeding due to leakage in weak capillaries causes HM. Larger than MA with irregular margin. They appear in dot and blot shape.
Hard Exudates (HE)—Lipoproteins and other fat proteins leaking through abnormal retinal vessels. It appears as white or yellowish-white deposits [24].
Cotton Wool Spots (CWS)—Soft exudates, occur due to occlusion of arteriole. It looks like fluffy bright white lesions.
Neovascularization (NV)—The growth of new abnormal blood vessels in the retina and the optic disk.

The early signs of DR include microaneurysms, hemorrhages, and future developing bright lesions such as exudates and cotton wool spots. Detecting these lesions is a challenging one because of variation in size, color, texture, and shape. However, manual inspection of retinal abnormalities diagnosis of early DR is quite an exciting task in medical image analysis. As a result, many kinds of research have been conducted on automatic identification of these abnormalities to reduce the burden on the ophthalmologists.

3.3 Stages of DR

There are two main stages of DR: non-proliferative diabetic retinopathy (NPDR) and proliferative diabetic retinopathy (PDR). NPDR occurs in early stage where the blood vessels start to bulge, outflow the blood makes the retina to get pathological lesions. NPDR has multiple levels which are explained in Table 4. PDR is an advanced severe stage that causes the growth of abnormal new blood vessels called neovascularization. New blood vessels bleed into the vitreous chamber and detach the retina by scar tissue formation which produces severe vision loss. Figure 11 shows the lesions for various stages.

Table 4 Clinical assessment of abnormalities for various DR stages

Stages of DR	Retinal anomalies
No DR	No abnormal signs
Mild NPDR	Few microaneurysms
Moderate NPDR	Microaneurysms and at least any one of the following: Hemorrhages Hard exudates Cotton wool spots
Severe NPDR	Any of the following: Intraretinal hemorrhages Venous beading or intraretinal Microvascular abnormalities
Proliferative DR	One or more of the following: Neovascularization Vitreous or preretinal hemorrhages

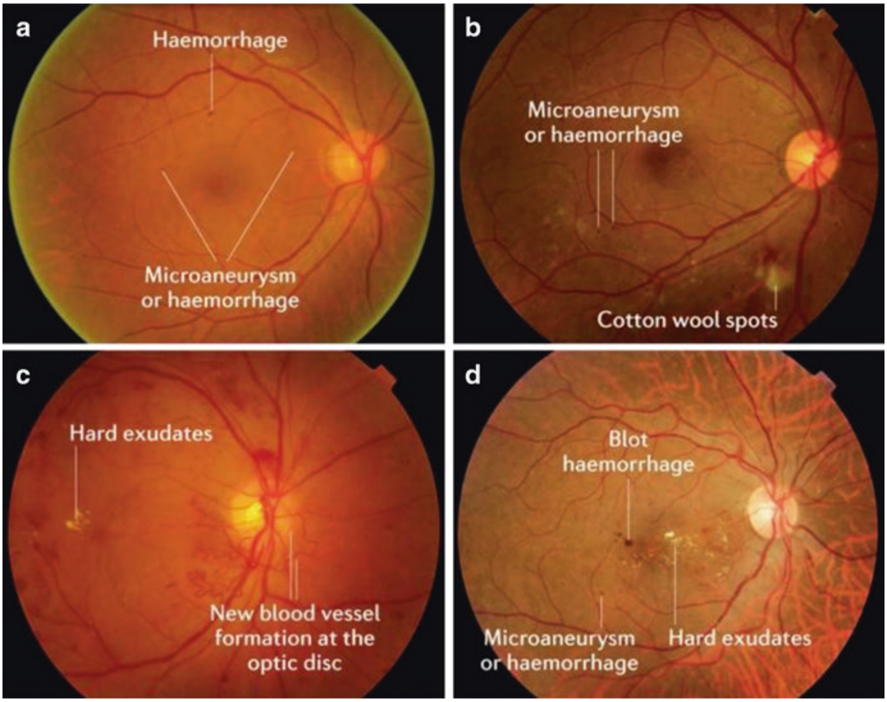


Fig. 11 (**a**) Mild NPDR, (**b**) moderate NPDR, (**c**) PDR, (**d**) macular edema [Source: Nature]

3.4 Datasets

The extensively used datasets for DR diagnosis are, namely, Kaggle, DIARETDB0, MESSIDOR, E-Ophtha, IDRID, STARE, DRIVE, and CHASEDB. Table 5 summarizes some of the datasets.

3.5 Early Prognosis of DR Using Retinal Images

Conventionally, researchers have used feature extractors and fed the features to classifiers like SVM and MLP. But the optimal selection of features is an essential requirement. DL has an excellent capacity to extract the features. This section presents the approaches used in diagnosing DR using DL. DR detection is characterized by lesion-level and image-level methods [25].

- **Lesion-based detection**—Number of lesions and their locations are detected in assessing the severity level of DR. It involves two steps, namely, lesion detection/segmentation and classification. Retinal lesions are extracted and the exact location of the lesion is confined. This is a challenging task because retinal

Table 5 Dataset available for DR research

Dataset name	No. of images	Disease category/purpose	Camera specifications
Kaggle/ EyePACS	35,000 images	DR No DR – Mild – Moderate – Severe – PDR	
IDRiD	516 images	DR, DME	
MESSIDOR	1200 images	DR	Color video 3CCD camera with 45° FOV
HRF	45 images	15 healthy 15 DR 15 glaucoma	Canon CR-1 fundus camera with 45° FOV
DIARETDB0	130 images	20 normal 110 DR	Fundus camera with 50° FOV
E-Ophtha	381 for MAs, 82 for exudates	148 MAs or small HA and 233 with no lesion, 47 with exudates and 35 with no lesion	
STARE	402 images	Retinal diseases—vessels extraction Optic nerve	
DRIVE	33 normal, 7 mild DR stage	Vessel segmentation	Canon CR5 non-mydriatic 3CCD camera with 45° FOV
CHASEDB	28 images	Vessels extraction	25° FOV

fundus images contain other objects with similar appearances, such as red dots and blood vessels.

- **Image-based detection—**Evaluates the image to detect whether there are signs of DR or not. Initial medical diagnosis starts with this image level using DL.

The presence of blood vessels and optic disc makes the lesion detection process more challenging because of DR lesions, and these two have similar attributes [25]. The general methodology for DR detection and classification includes the steps of pre-processing, segmentation, feature extraction, and appropriate classification method to detect the severity level of DR as shown in Fig. 12. Unlike manual feature-based approaches, DL methods integrate all of the phases into an integrated context and automatically learn the features and train the model. Deep architectures are able to learn hierarchical features and achieve excellence in medical diagnosis. The complex features are recognized by having a greater number of hidden layers in the network. Automatic feature extraction and limited pre-processing are the main advantages of DL. Classification learning algorithms can be mainly grouped into two categories, namely, supervised and unsupervised learning. Mostly, supervised methods are used in existing studies for screening DR diagnosis [24].

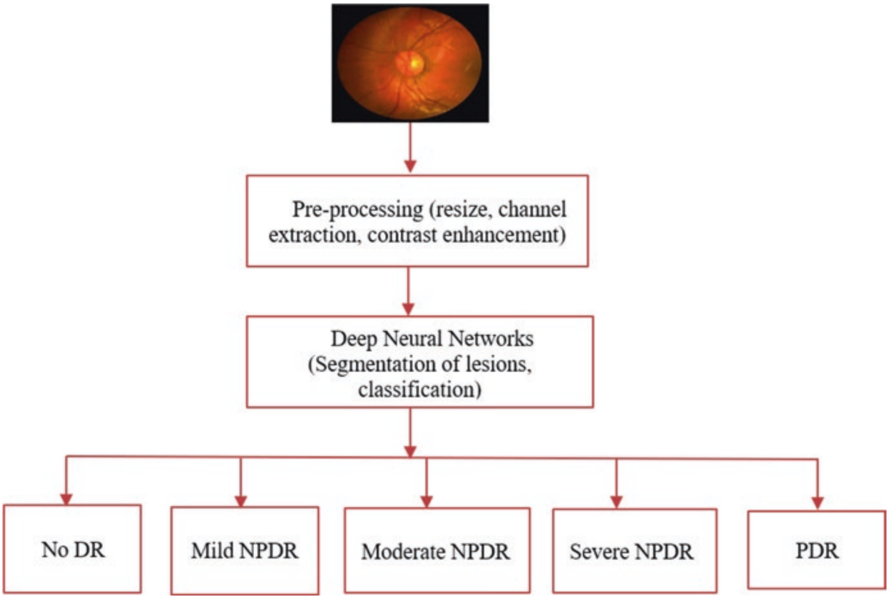


Fig. 12 Stages in DR diagnosis

3.5.1 Pre-processing of Retinal Images

The reliable screening system depends on the quality of the fundus image. However, clinical studies emphasized 3–12% of retinal images having low quality [26]. Retinal images were captured in different conditions like dissimilar cameras, non-uniform illumination, variations in light reflection, poor contrast, and insufficient pupil dilation. These may affect the accuracy of the diagnosis methods. Image pre-processing techniques improve the diagnostic probability for visual and automatic detection. Pre-processing of retinal image analysis typically covers image resize, noise removal, channel extraction, and contrast enhancement.

3.5.1.1 Extraction of Green Channel

The retinal image exhibits low contrast due to the fluctuating nature of imaging environments. Most of the retinal lesions and blood vessels are perceptible in the green channel compared to red and blue channels as shown in Fig. 13. The green channel is extracted and enhanced to improve the contrast.

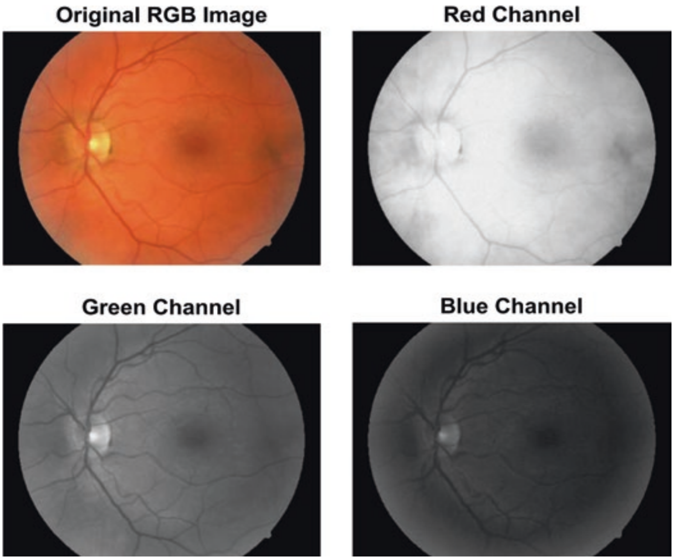
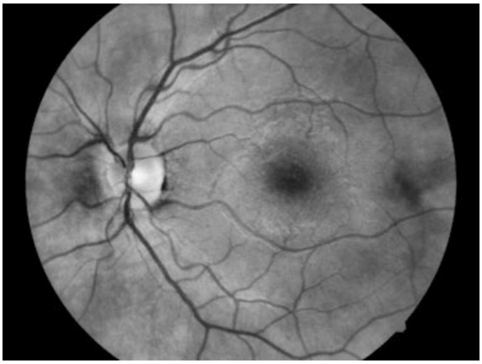


Fig. 13 Individual RGB channels of retinal images

Fig. 14 Contrast-enhanced image



3.5.1.2 Contrast Enhancement

Contrast limited adaptive histogram equalization (CLAHE) is a variation of histogram equalization which helps to enhance the local contrast between background and blood vessels dynamically. CLAHE divides the retinal image into a nonoverlapping region called tiles and applies the histogram distribution for each tile with a clip limit. After applying CLAHE, the segmentation of blood vessels in fundus images has improved for DR disease identification as shown in Fig. 14. Filtering techniques may be used to smooth the background noise.

3.5.2 Segmentation of Retinal Pathologies

Structures like microaneurysms, hemorrhages, hard exudates, and cotton wool spots in fundus images are essential to the diagnosis and screening of DR. As already discussed, the attributes of these features have similarity in appearance, and lesion detection is a more challenging part in NPDR and PDR disease identification given in Fig. 15. Recently, many deep architectures like CNN, deep CNN, and transfer learning have been widely used in extracting the features. In this section, we discuss the method adopted to create the training data for learning and DL approaches used for lesion detection for further analysis of DR.

Microaneurysm—Microaneurysms are the early pathology developed in the mild stage of diabetic retinopathy (DR), and its precise detection from fundus image is a challenging task. Since the occurrence of MA looks like a microscopic change in pixel level out of total volume of pixels and variation in size of MAs, accurate diagnosis approaches are required. CNN has the ability to extract deep features, and studies have shown that CNN forms the basis of all developed algorithms.

The extensively used methods of segmenting MA have been discussed as follows:

1. Image patches are generated from retinal images which contain MAs and non-MA lesion, features are extracted by training a network using MA and non-MA patches, and binary classification is performed. But this method requires the labeled annotated images for training patches [27]. For training, deep CNN architectures have been widely adopted in researches. Also, the autoencoder model is developed to learn high-level features of MA and fed into the Softmax classifier [28].
2. Every pixel of the image is classified as MA or non-MA using deep neural networks [29].

Also, both the patch-based and image-based methods have been combined, i.e., patches with annotated labels are used for training and trained CNN was used to test the images in pixel-wise and identify the locations of the pathological signs with probability maps [30]. But the challenge lies in labeled data and making the training set balanced in both class MA and non-MA. Additionally, clinical grading has been adopted to improve deep model performance [31]. Moreover,

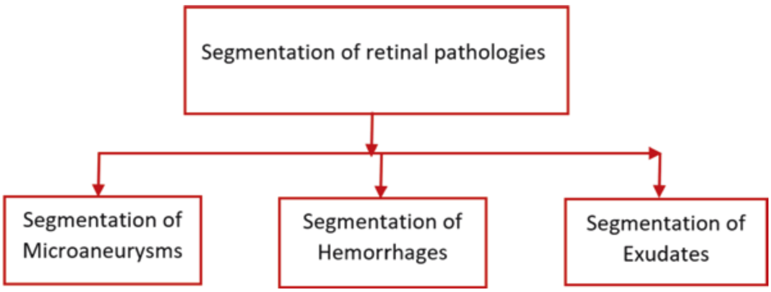


Fig. 15 Segmentation of retinal abnormalities

combining handcrafted features and deep features learned from CNN has worked on the detection of MA.

Hemorrhages (HM)—Hemorrhages occur due to leakage of blood from ruptured blood vessels. Both patch- and image-based investigations can be used in detecting hemorrhages. All the process involved in detecting MA is applicable to HA [32].

Hard Exudates (HE)—Hard exudates are formed by the deposition of lipids and lipoproteins located in the outer layer of the retina, with whitish spots or yellowish-white appearance. The process of detecting HE is similar to MA. Image patches of size 32 × 32 are generated by using the ground truth into two classes, hard exudate and background. Image patches are given for training an eight-layer CNN and achieved 99.4% training accuracy and 98.6% testing accuracy [33]. The main challenge lies in predicting edge pixels. The image processing techniques such as green channel extraction and contrast enhancement can improve the accuracy along with supervised algorithms to detect exudates [34].

Future research can be extended in two approaches: The first approach is to vary the size of the image patch. Then we can find the effect of image patch on the accuracy of prediction. The second approach is to use an ensemble of CNN architectures.

3.5.3 Deep Learning Architectures to Diagnose DR

Deep learning has shown capable improvements in retinal image analysis for the detection of eye and systemic diseases. Though such techniques require high training time and extensive data, the development of transfer learning has been chosen as a diverse method to optimize the process [35]. In this section, some of the researches which used CNN in DR detection are discussed.

Convolutional Neural Networks (CNN)—With the increasing popularity of CNN in medical image analysis, CNN can identify features such as microaneurysms, exudate, and hemorrhages in the retina and provide a diagnosis automatically. CNN models comprised of the convolutional layer, pooling layer, and dense layer. Activation functions provide nonlinearity to the model, and backpropagation is used to update the weights and learn the model. Batch normalization and dropout are commonly adopted to achieve faster convergence. CNN requires large datasets for training to avoid overfitting, but the ophthalmic diagnosis has smaller datasets. Transfer learning and data augmentation help to train the model using large datasets, followed by fine-tuning on the test dataset. Also, ensemble learning involves training many CNN models independently and predicts the result.

The paper [36] implemented a CNN architecture to identify the features in the retina and achieved an accuracy of 75% on 5000 validation images. Transfer learning method with InceptionNet V3 network achieved an accuracy of 90.9% [37]. Pre-trained models like ResNet50, Xception Nets, DenseNets, and VGG are used and achieved accuracy of 81.3% [38]. Transfer learning method with three tasks, namely, classification, regression, and ordinal regression, was performed. Each task is trained based on features extracted with CNN that produced a sensitivity of 0.99 [39].

Table 6 Evaluation metrics for DR

Metrics	Description	Formula
Accuracy	Ratio of the correctly classified instances over the total number of instances	$\frac{TN+TP}{TP+TN+FP+FN}$
Sensitivity	Measures the fraction of correctly classified positive diagnosis	$\frac{TP}{TP+FN}$
Specificity	Measures the fraction of correctly classified negative diagnosis	$\frac{TN}{TN+FP}$
Area under the curve (AUC)	Degree of separability of different classes	–

The ensembles of deep convolution neural network (CNN) models such as Resnet50, Inceptionv3, Xception, Dense121, and Dense169 are combined to improve the classification for different stages of DR on Kaggle dataset. The results were better compared to the individual model [40].

Nowadays, researchers are developing a smartphone-based early DR detection system. Early detection of DR generally involves eyes to be dilated. Retinal images were acquired using the smartphone "Remidio camera" and diagnosed using artificial intelligence techniques [3] (Table 6).

3.5.4 Research Ideas

Even though deep learning has grown over state-of-the-art methods, training the network and fine-tuning the parameters still is a challenging area. Problems such as poor image quality, differences in the size of lesions, and the closeness of some lesions to the vessels have caused many detection algorithms to provide low-accuracy results. Moreover, MA which is very close to blood vessels is very difficult to be segmented. Blood vessel segmentation requires precise consideration because thin vessels also have a contribution in disease screening.

The automated diagnosis methods can be used as a screening tool to support the clinicians. Non-mydriatic detection system may be developed in the future research. Clinicians can also diagnose in an accurate way after performing preliminary steps using an automated system.

4 Conclusion

Methods of efficient waste segregation for developing smart bins have been reviewed.

The data acquisition system might be useful in capturing real-time images. This paper discussed the identification of waste from images using deep learning. Mostly, CNN has been widely adopted by the researchers to classify the waste items. The concept of transfer learning is involved in a small number of datasets to improve performance accuracy. Since the real-time images have multiple objects, the idea of identifying multiple objects in a single image is yet to be recognized and classified. The categories of the dataset used may be extended in the future. Research openings can reduce the response time in determining the waste after being thrown away.

Retinal image analysis through deep learning is evolving to make the screening process automated. Deep learning techniques can be applied for the segmentation of retinal lesions to detect the stages of DR. Deep learning has the capability of extracting more complex features than those extracted by traditional methods. Deep neural networks (DNN) based automated diagnosis model can be used in the clinical screening process which replaces the skilled ophthalmologists and reduces the time. Although much researches have been conducted to implement the same in real time, the epitome of the method is yet to be observed.

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